Spring Project in an Hour

Build your Snowflake AI Agent for Conversational Analytics

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Project Story: Connecting Patient Records to Medical Research

When treating a patient, the information a doctor needs is often split between two different worlds:

- **The Patient's File:** This contains structured history like specific **medications, procedures, immunizations, and allergies**. It's a detailed record of what has happened to a person over time.
- **The Library of Research:** This contains millions of full-text articles filled with the latest medical breakthroughs and clinical trends.

In this tutorial, you will build an AI assistant that understands both at once. Instead of searching these sources separately, you can ask a question in plain English. The AI will look at a patient's actual records—like their **conditions or procedures**—and compare them to global research to give you a clear, cited answer.

Project Data: Healthcare Clinical + Claim Data and Biomedical Research Data

We will access data from two different sources in Snowflake Marketplace.

- **Structured Data:** The synthetic healthcare data – clinical and claims in "Snowflake Public Data (Free)"

This synthetic healthcare dataset includes comprehensive clinical & claims Data: The dataset includes patients, conditions, encounters, medications, procedures, observations, allergies, immunizations, and related claims and provider tables. Currently, the database includes more than: 1.4M Patients, 65M Encounters, 124M Claims, and 887M Claim Line Details

- **Unstructured Data:** The PubMed Biomedical Research Corpus in "Snowflake Public Data (Free)" .

This listing provides a Cortex Knowledge Extension allowing you to ask questions of biomedical and scientific research included in the PMC Open Access Subset and get back summarized, cited answers in your AI assistant and agentic applications built on Snowflake Cortex AI.

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Prerequisites:

- A Snowflake account (educators can apply for a free Snowflake account from the Snowflake Academia program and share with students).
- Basic knowledge of Snowflake UI, concepts, and objects (database, schema, table, virtual warehouse, etc)

What You Will Learn

- **Cortex Analyst tool:** a fully managed agentic system that uses a Semantic Layer to turn natural language into precision SQL to query structured data like the healthcare dataset.
- **Cortex Search tool:** a hybrid (vector + keyword) search engine that provides the "RAG" (Retrieval-Augmented Generation) capabilities to read unstructured data like the medical journals.
- **Cortex Agent (Orchestration):** the "brain" that routes user questions to the correct tool.

You will learn how to create these building blocks for Snowflake Intelligence.

What You Will Build

An Enterprise Intelligence Agent - Snowflake Intelligence - that can respond to questions by reasoning over both structured data (healthcare data) and unstructured data (medical research corpus)

Cortex Analyst

This tool enables the agent to query structured data in Snowflake by generating SQL. It relies on semantic views, which are mappings between business concepts (e.g., "patient name," "immunizations") and the underlying tables and columns in your Snowflake account. This abstraction helps the LLM understand how to query your data effectively, even if your tables have complex or arbitrary naming conventions.

Access the structured data

- In Snowsight, on the left hand navigation menu, select **Marketplace**
- Search for Synthetic Healthcare
- Select **Synthetic Healthcare Data – Clinical and Claims**
- Select **Get**
- Select **Get** (again)

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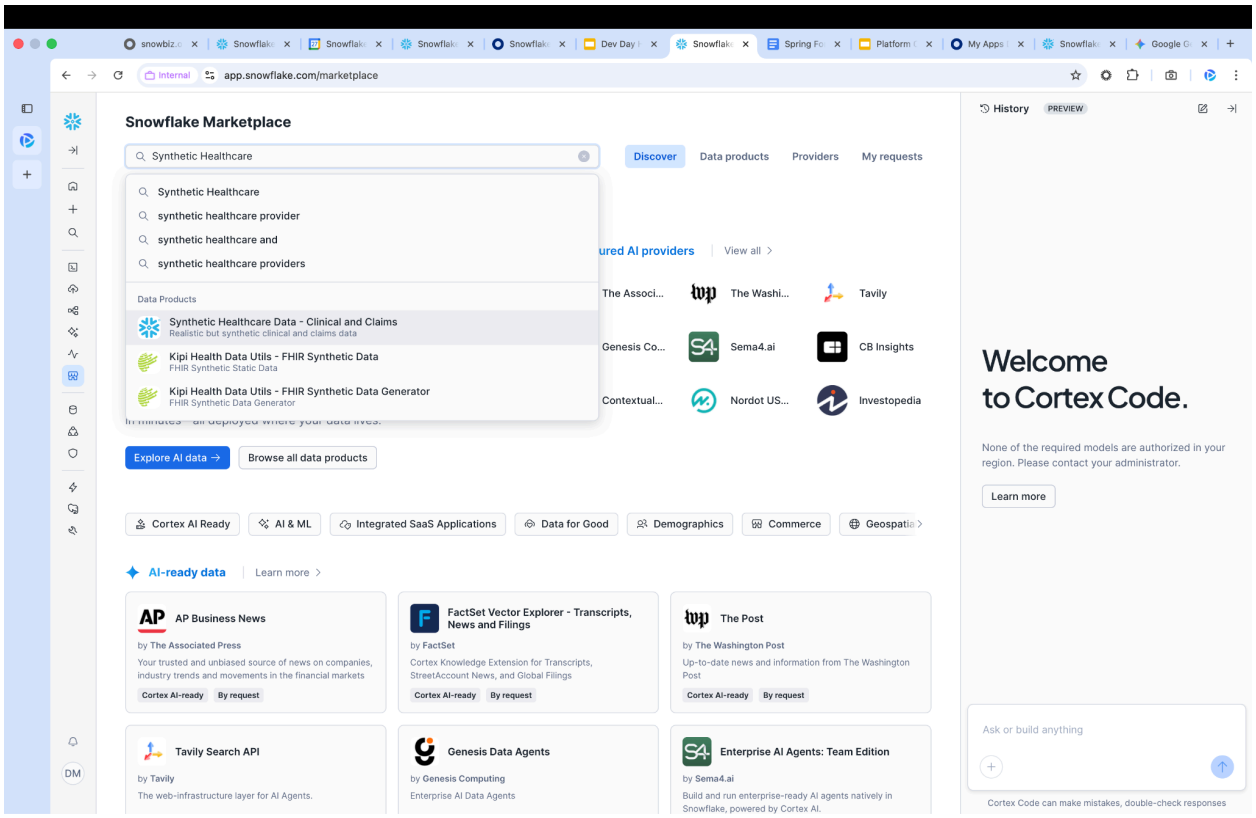
If you encounter the "dataset already imported" error, it means another user has already “gotten” or “imported” the list and it becomes available in this account for you to use.

When a user in a Snowflake account accesses a dataset from the Snowflake Marketplace, the data is not copied or moved; instead, a shared database is created at the account level and available for any authorized user to query.



Create The Semantic Layer

The Semantic layer turns natural language into precision SQL to query structured data like the healthcare dataset.

- Create or open a SQL worksheet and run this SQL statement “**CREATE DATABASE [login]_PATIENT_INSIGHTS_DB**”

  This database will be used to store all the objects that will be created in the rest of the tutorial.

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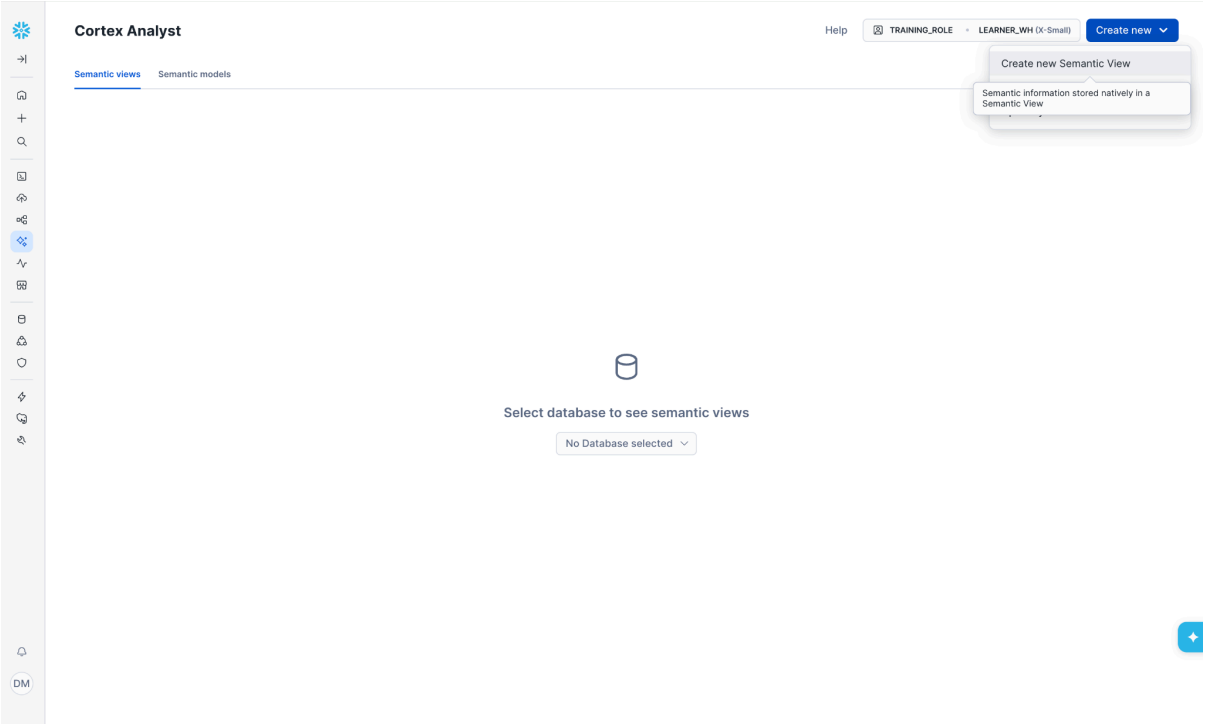
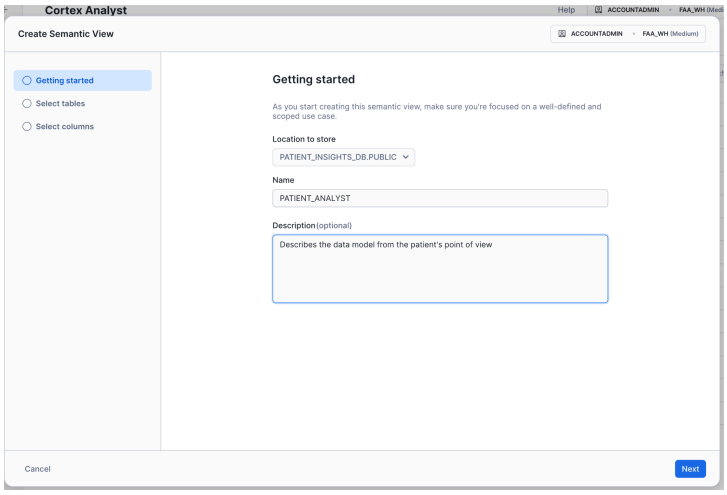
- **Create the Semantic View:**
  - Hover over **AI/ML** on the left-hand navigation
  - Navigate to **Cortex AI** and select **Analyst**.
  - Select **Create New** > **Semantic View**.



- In your **Location to store,** enter **[login]_PATIENT_INSIGHTS_DB.PUBLIC**.
- In the **Name** field, enter **PATIENT_ANALYST**
- In the **Description**, enter "**Describes the data model from the patient's point of view.**"

This metadata helps the agent understand the scope of its knowledge.



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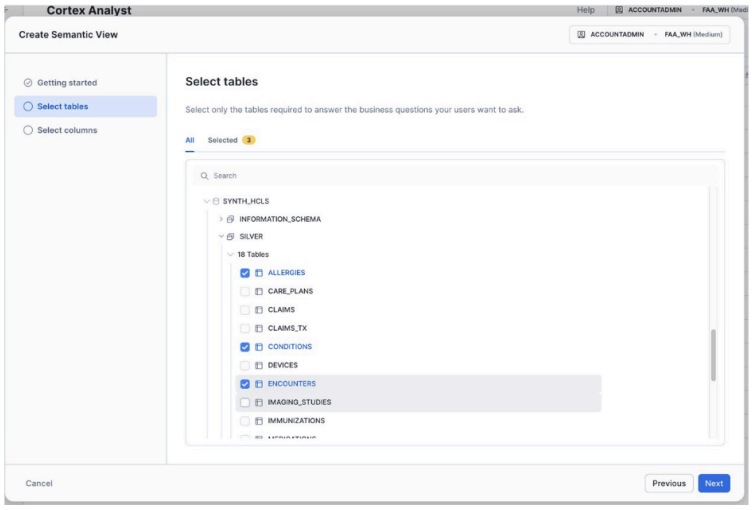
- ## Select Tables for the Semantic view:

  - Locate the **SYNTHETIC_HEALTHCARE_DATA_CLINICAL_AND_CLAIMS** database
  - Select **Silver** and then select **18 tables**
  - Select these 6 tables (**Patient, Conditions, Encounters, Medications, Allergies, Procedures**) by clicking the white square
  - Select Next



- ## Select Columns for the Semantic View:

  - For each table, click on the white square. This selects all of the columns in the table.
  - Ensure you check "**Add sample values**" and "**Add descriptions**" This allows Snowflake's AI to "look ahead" at your data to suggest descriptions, ensuring the LLM understands exactly what each column represents.
  - Select **Create and Save**

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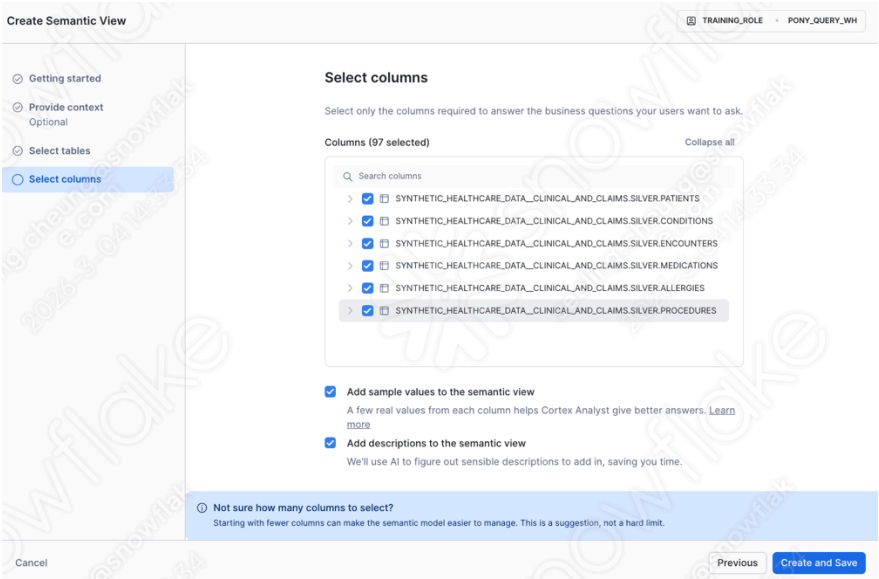

- **Set Primary Keys for the Semantic View:**
  - For each table, select the **Edit** button on the right side of the center window.
  - Select **+ Primary Key**
  - Select the field that looks like TABLE_NAME_ID. For example, if the table is PATIENTS, then look for the field patients_id to add as primary key
  - Select **Save**.
  - You will need to repeat this process for all six tables.

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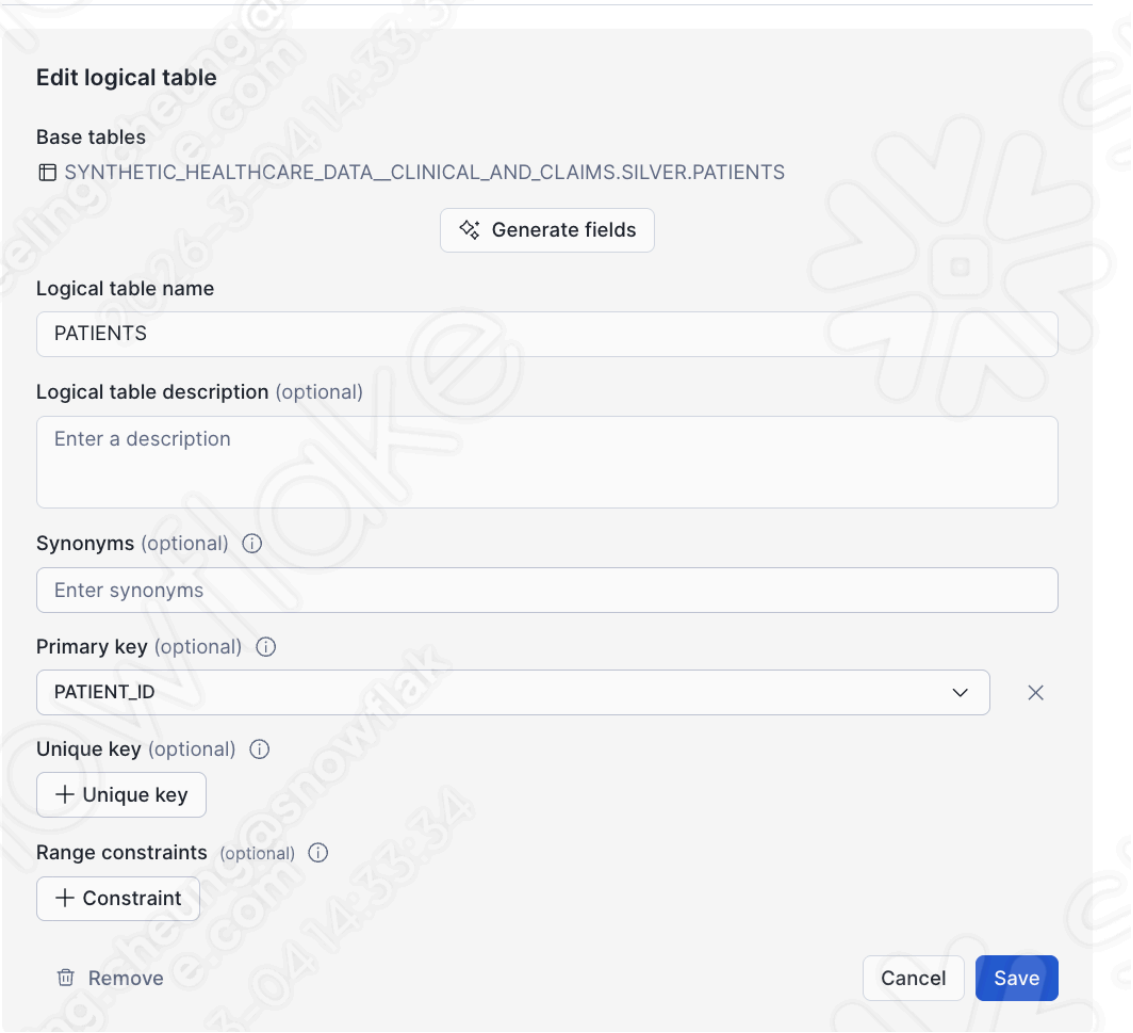

- **Set Relationship for the Semantic View:**
  - In the **Relationships** area of your Semantic View, select the **+ sign** on the right side.
  - Enter **PATIENT_ALLERGY** in the Relationship name field.
  - Select **ALLERGIES** as the Left Table. Select **PATIENTS** as the Right Table.
  - Select **PATIENT_ID** for both the Left and Right column fields.
  - Select **Add**.

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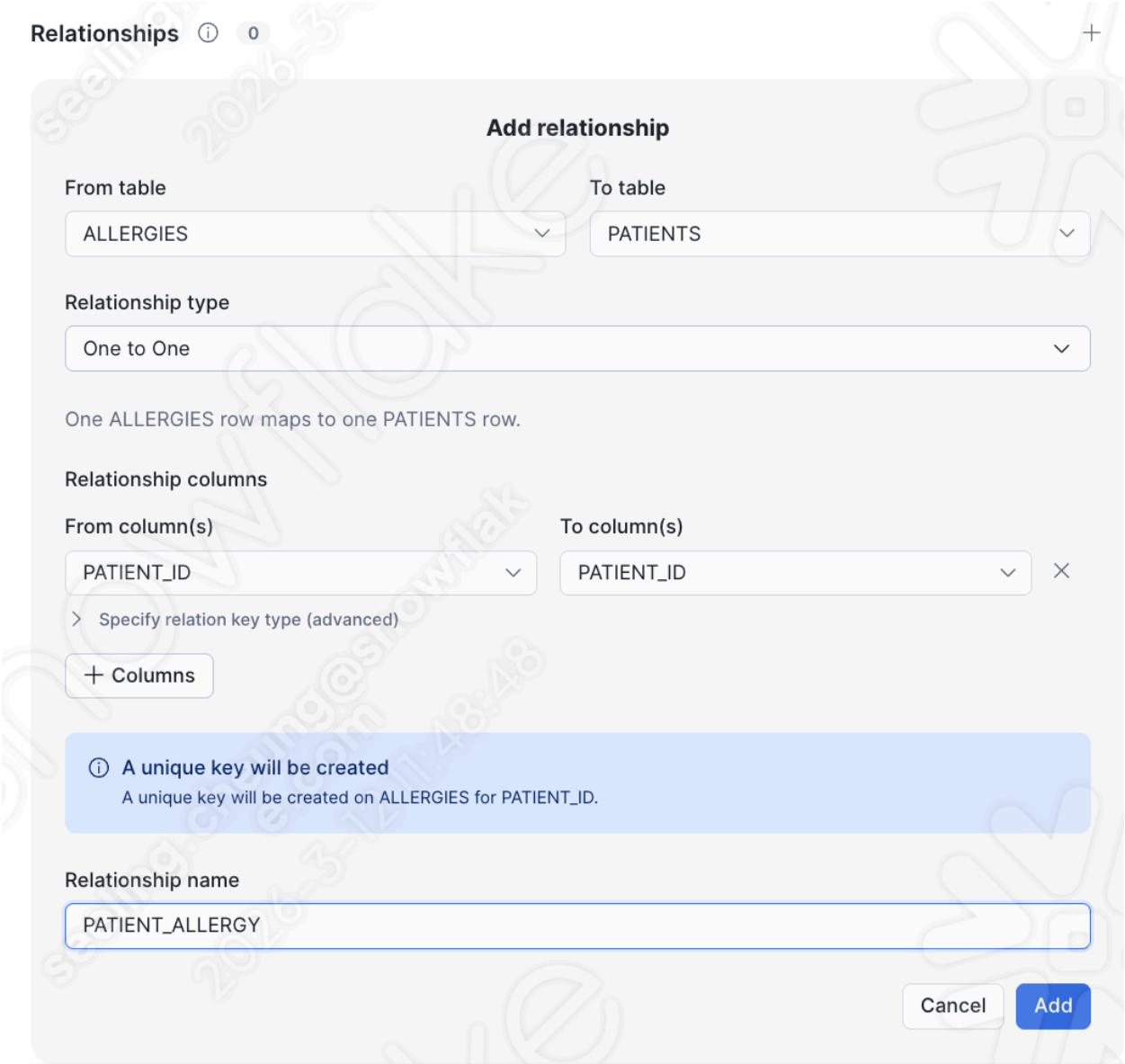

You have now completed the steps for Cortex Analyst!

Cortex Search

This tool allows the agent to search and retrieve information from unstructured text data, such as customer support tickets, Slack conversations, or contracts. It leverages Cortex Search to index and query these text "chunks," enabling the agent to perform Retrieval Augmented Generation (RAG).

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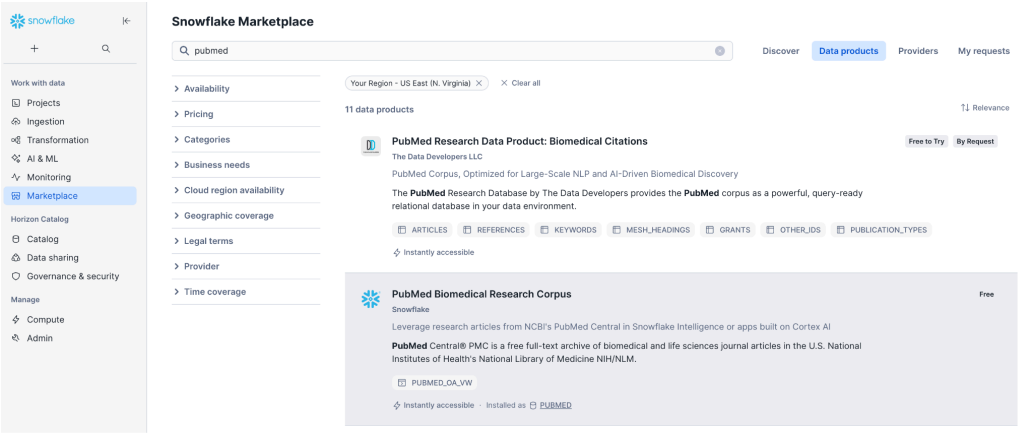
Access the unstructured data

- Select **Marketplace**
- Search for **PubMed**
- Select **PubMed Biomedical Research Corpus**.
- Select **Get**
- Select **Get** (again)



If you encounter the "dataset already imported" error, it means another user has already “gotten” or “imported” the list and it becomes available in this account for you to use.

Cortex Search service

After getting the listing PubMed Biomedical Research Corpus, you will find a new Cortex Search Service shared with your account.

- Select **PubMed Biomedical Research Corpus** from **Catalog->Database Explorer**
- Select **OA_COMM** schema
- Select **Cortex Search Services**
- You will find the service named **PUBMED_OA_CKE_SEARCH_SERVICE**

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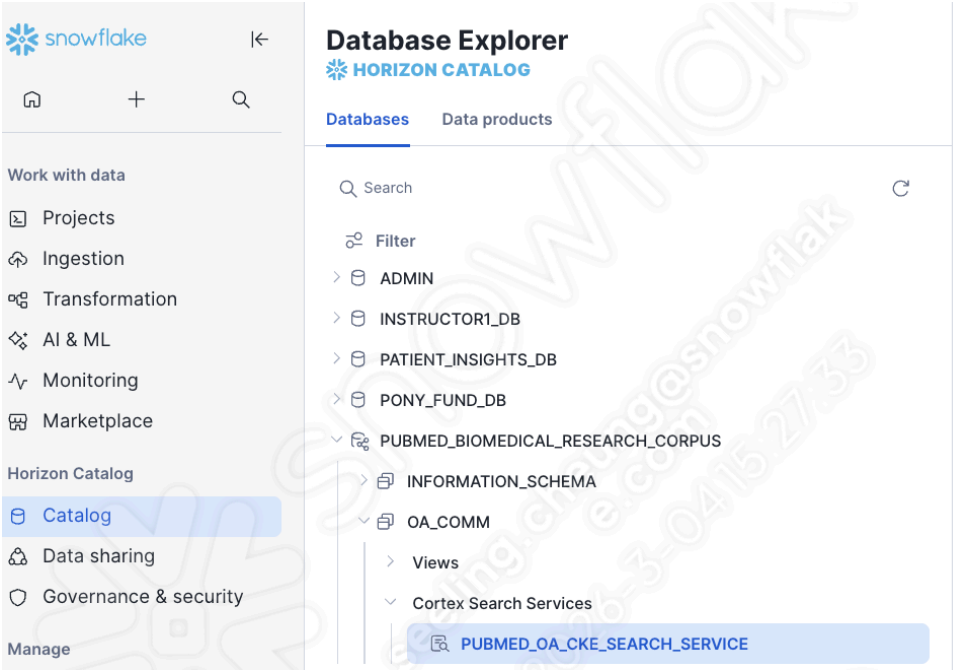

We will connect this service to the agent we will build in the next step. This will allow us to ask questions of biomedical and scientific research included in the PMC Open Access Subset and get back summarized, cited answers.

Agent Creation & Orchestration

An agent is an intelligent entity within Snowflake Intelligence that acts on behalf of the user. Agent unifies standalone AI tools inside Snowflake like Cortex Analyst and Cortex Search under one API, provides reasoning, picks right tools for the task and gives answers.

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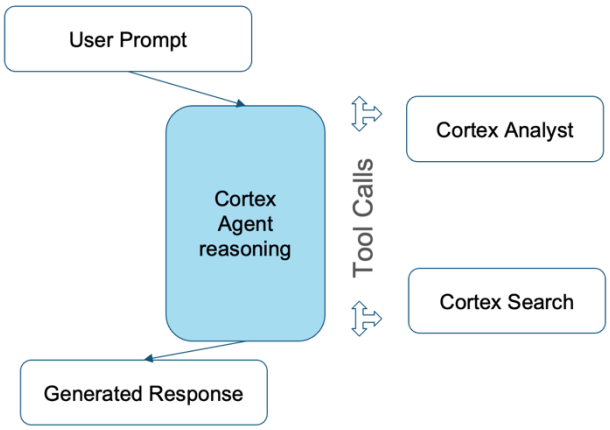
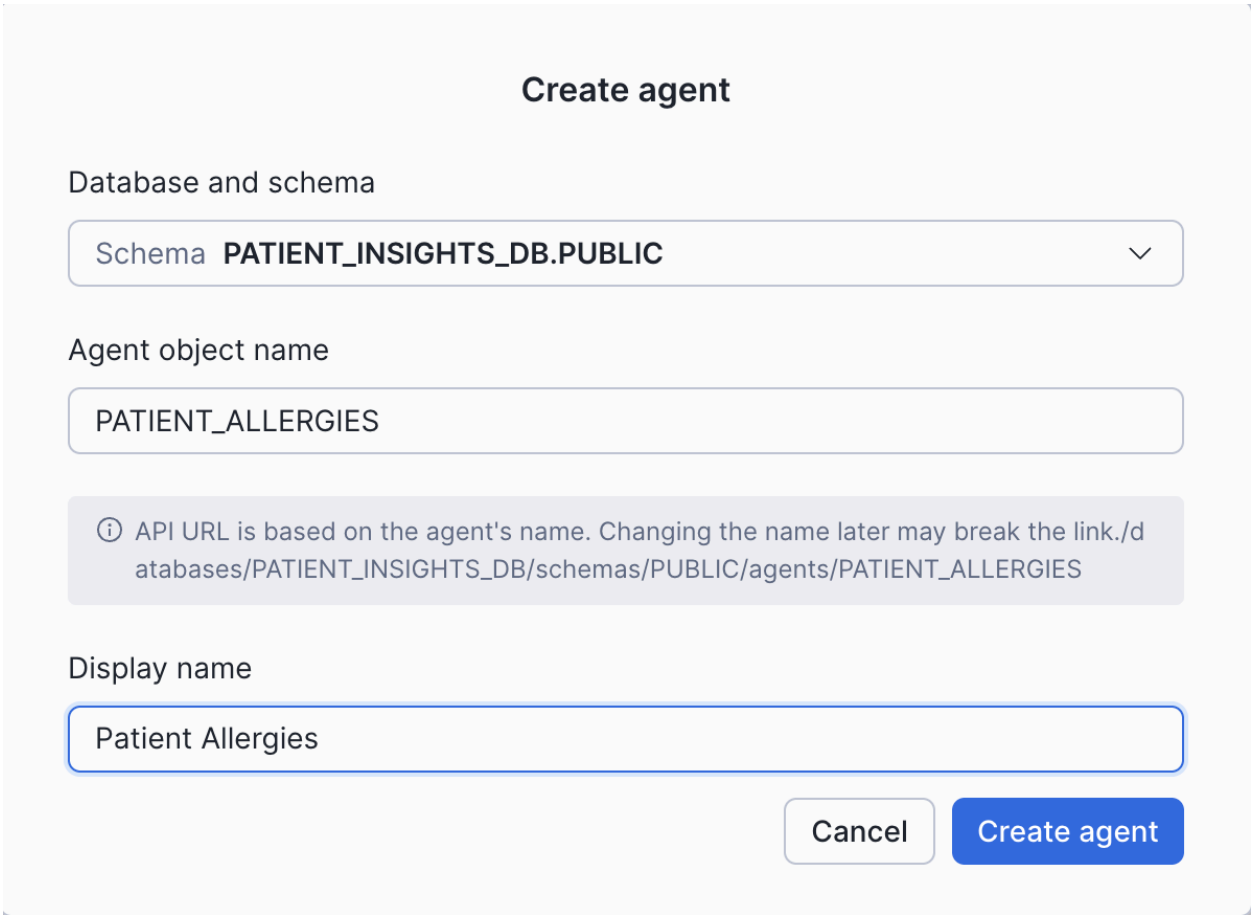


Create the agent

- Hover over **AI & ML**
- Select **Agents**
- Select **Create Agent**. Name it **PATIENT_ALLERGIES**



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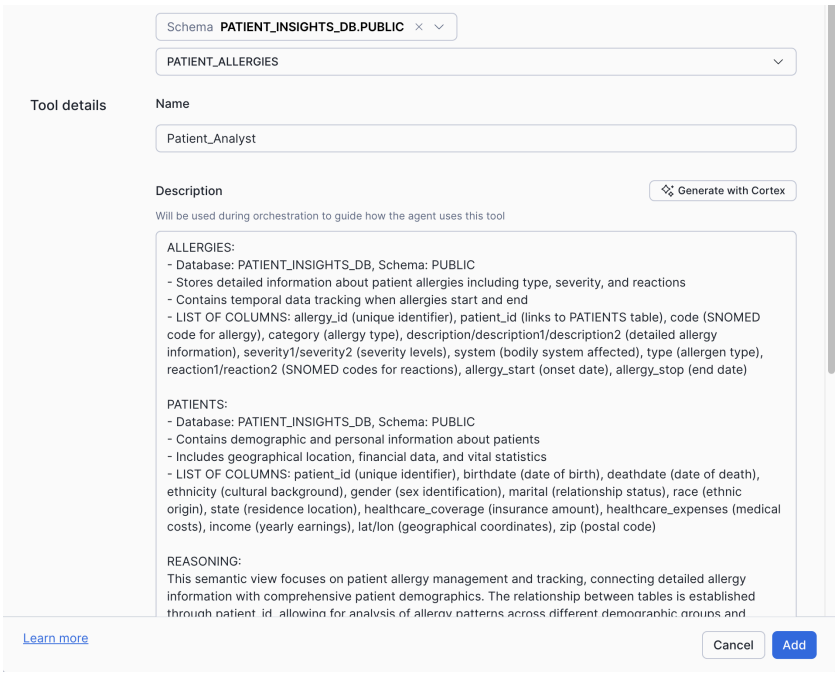
Add Tools

Tools are the capabilities an agent can use to accomplish a task. Think of them as the agent's skillset and note that you can add one or more of each of the following tools.

Add the analyst tool to the agent (adding the semantic view)

- Select the agent **PATIENT_ALLERGIES**
- Select **EDIT**
- Select the tab **Tools**
- Select **+Add** next to **Cortex Analyst**
- Select database **PATIENT_INSIGHTS_DB** and schema **PUBLIC**
- Select **PATIENT_ANALYST** semantic view
- In the Name field, enter **Patient_Analyst**
- Select **Generate with Cortex**
- Select **Add**



Add the search tool to the agent (adding the semantic view)

- Select **+Add** next to **Cortex Search Services**
- Select database **PUBMED_BIOMEDICAL_RESEARCH_CORPUS** and schema **OA_COMM**

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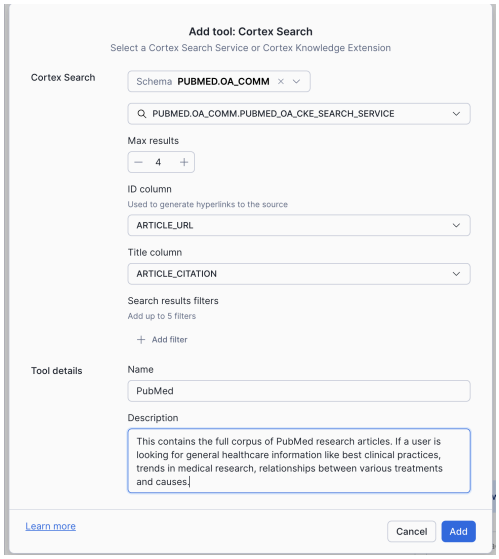
- Select **PUBMED_BIOMEDICAL_RESEARCH_CORPUS.OA_COMM.PUBMED_OA_CKE_SEARCH_SERVICE**
- Set **Max results** to 4
- For the **ID** column, select **ARTICLE_URL**
- For the **TITLE** column, select **ARTICLE_CITATION**
- For the **Name**, enter **PubMed**
- For the **Description**, enter **"This contains the full corpus of PubMed research articles. If a user is looking for general healthcare information like best clinical practices, trends in medical research, relationships between various treatments and causes."**
- Select **Add**
- Select **Save**



Because Cortex Analyst supports **Multi-Turn Conversation,** your agent will remember context. If you ask about "life-threatening allergies" and then follow up with "What about latex?", the agent knows you are still talking about lethal risks.

Test Your Agent

- In the Agent screen, select your agent **PATIENT_ALLERGIES**
- In the agent window, enter your questions

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Some sample questions

Question 1 : Which allergies are potentially life threatening?

Which allergies are potentially life threatening?

✓ **Thinking completed** Show Details

Several types of allergies can be potentially life-threatening by causing anaphylaxis, a severe and rapid-onset allergic reaction:

Food allergies are among the most common life-threatening allergies, with approximately 200 to 250 million individuals worldwide experiencing food allergies. In infants, egg and cow's milk are the most common foods causing anaphylaxis.

Drug/medication allergies can also cause life-threatening reactions. These hypersensitivity reactions to medications can be deadly in some people.

Insect sting allergies can trigger life-threatening anaphylaxis, with sting-induced anaphylaxis being a recognized dangerous allergic condition.

The life-threatening nature of these allergies stems from anaphylaxis, which involves respiratory or cardiovascular system compromise. Symptoms can include severe bronchospasm, low blood pressure, rapid or slow heart rate, difficulty breathing, and loss of consciousness. Anaphylactic reactions typically occur within seconds to minutes after exposure, though some symptoms may appear up to 2 hours later.

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Question 2: What are some ways to prevent lethal exposure to latex?

What are some ways to prevent lethal exposure to latex?

✓ **Thinking completed** Show Details

Preventing lethal exposure to latex involves multiple prevention strategies:

Primary Prevention (Reducing Exposure)

For High-Risk Individuals:

- Identify at-risk populations early, including children with spina bifida, those undergoing repeated surgeries, or those with urogenital anomalies
- Use latex-allergen-free equipment in operating rooms for high-risk patients, which has been shown to reduce latex allergic sensitization

Environmental Measures:

- Create "latex-safe" environments through legislative restrictions on latex glove use in hospitals
- Use alternatives to latex products, such as synthetic elastomers or Yulex gloves made from Guayule
- Mandate labeling of all products containing latex for easy identification and avoidance
- Use products with reduced allergenic latex content through deproteinization and purification processes

Secondary Prevention (Early Diagnosis)

- Promptly refer patients with suspected latex allergy for allergy evaluation
- Workers should report any suspected allergic reactions immediately for timely diagnostic testing

Tertiary Prevention (Managing Diagnosed Cases)

Individual Protection:

- Provide patients with an action plan for managing allergic reactions
- Prescribe adrenaline auto-injectors to prevent anaphylaxis
- Use identification bracelets for pediatric patients with latex allergy

Institutional Measures:

- Alert school staff about students with latex allergies and provide individualized action plans
- Remove latex-containing products from schools (balloons, erasers, school materials)
- Relocate affected workers to latex-safe work areas or adjust their roles

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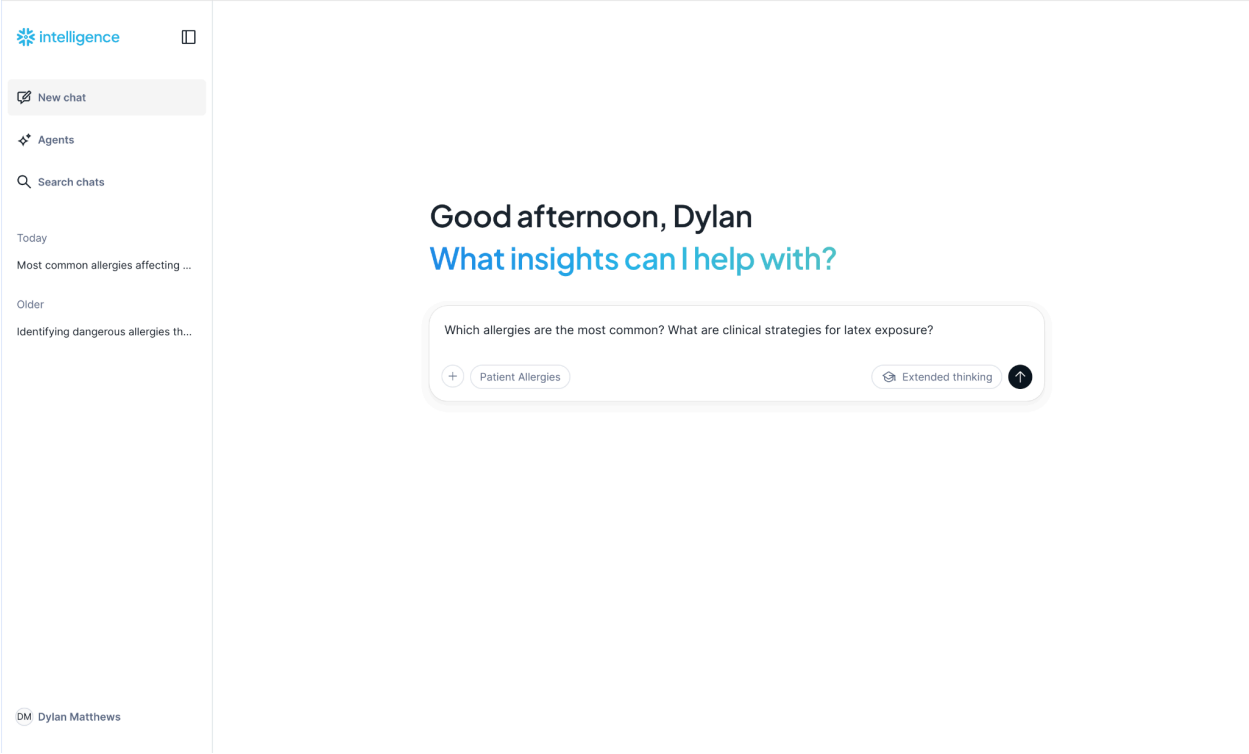
Snowflake Intelligence

Create Snowflake Intelligence

- Hover over **AI & ML**
- Select **Snowflake Intelligence**

A new window will appear for accessing Snowflake Intelligence.

Now, let's ask questions.



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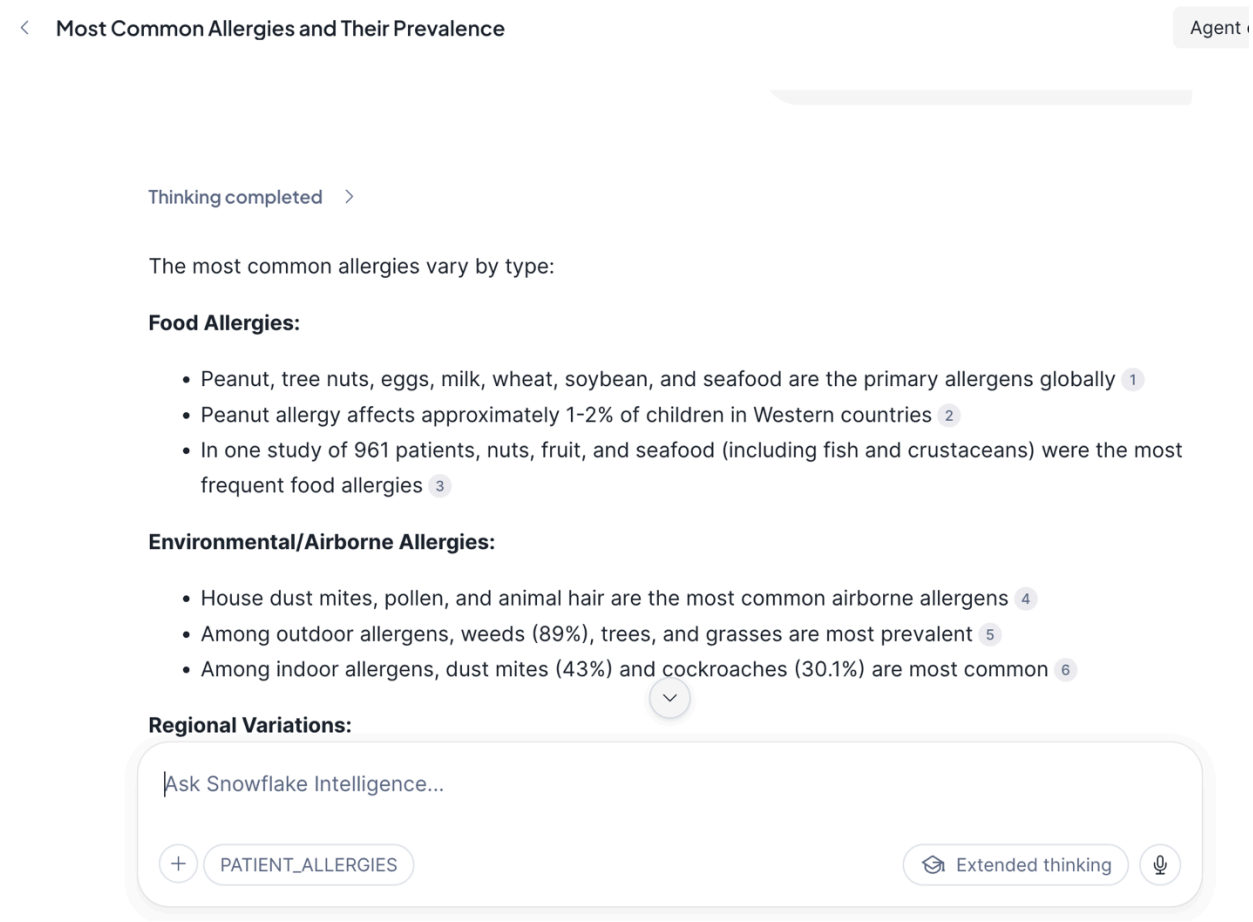
Ask a Question on Structured Data: "Which allergies are the most common?"

Cortex Analyst will generate a SQL query, execute it in your warehouse, and return the answer.



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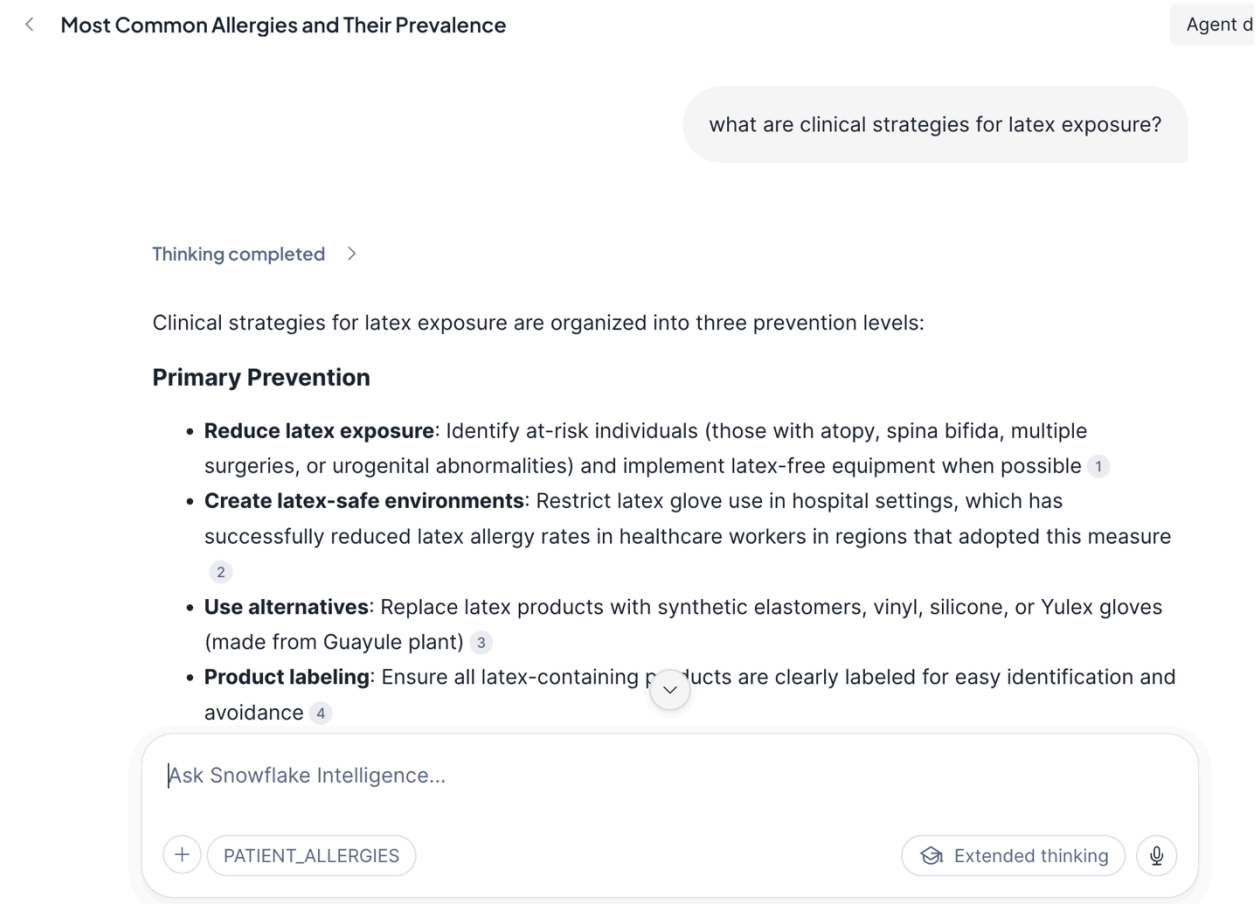
Ask a Question on Unstructured Data: "What are clinical strategies for latex exposure?"

The agent pivots to Cortex Search, retrieves the latest PubMed abstracts, and summarizes them.



Conclusion And Resources

Congratulations! You've successfully built an enterprise intelligence agent - Snowflake Intelligence - that can respond to questions by reasoning over both structured and unstructured data.

What You Learned

You've learned how to create the fundamental building blocks for Snowflake Intelligence. This agent is now a powerful asset, capable of combining analysis and automated tasks right from a single chat interface.

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Related Resources

Snowflake Intelligence and Documentation

Snowflake Cortex Analyst Documentation

Snowflake Cortex Search Documentation

Snowflake Marketplace and Documentation

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